

jQuery - jQuery Mobile GSoC 2015 Proposal

Upgrade and Refactor Test Suite

[Latest Version](#)

Name and Contact Information

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University: Indian Institute of Technology, Roorkee

Field of Study: Electrical Engineering

Expected Graduation Date: May, 2016

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Location: **Roorkee, Uttarakhand, India**

Typical Working Hours:

10:00- 13:00, 15:30-19:00, 22:00-03:00 (IST) 4:30- 7:30, 9:30-1:30, 16:30- 21:30 (UTC)

Other Interests: Security, Networking, Gaming

Coding Skills:

- [Github Profile](#) (Total 507 stars)
- [Project Page](#)

- Joint Secretary of [SDSLabs](#)
- GSoC 2014 Participant with Mediawiki ([Project](#))
- Winter 2014 Intern with [Wingify](#)
- [About Me Page](#)
- [Pull Requests to jQuery Mobile](#)
- Regular Contributor to Open Source Organizations including GNOME, Moodle, Mediawiki.
- Knows PHP (Laravel), JavaScript (Angular and Backbone), Python, Ruby, Go, Node.js, Bash, Clojure, Lua, C++ as depicted by Github Profile.

Proposal

- **Public Url**
- **Discussions:** [\[1\]](#) [\[2\]](#) [\[3\]](#) [\[4\]](#) [\[5\]](#)
- [Prototype](#) for this project

Synopsis

Project aims to completely refactor and upgrade the current test suite of jQuery Mobile so as to ensure maximum coverage on all the supported platforms. During the project, we will test various testing libraries primarily [Intern](#) to improve current tests and ensure best quality code.

Detailed Description

Need

Testing is important as each of us make mistakes. So the concern is that previous available functionality doesn't break as we move forward. Thus each and every part of code must be tested to guarantee its functionality in future.

As new testing libraries emerge everyday, tests are getting more easy to write and read. An obvious need is felt to constantly upgrade and refactor current test suite using available better tools and libraries. Cross platform CI and webdriver integration are supported in many of them by default. In our case, we will use **Intern** to upgrade our tests because it supports all the features needed out of the box.

Mentors

1. Alexander Schmitz ([arschmitz](#))
2. Sarah Frisk ([sfrisk](#))

Proposed Flow:

- **Integrate with Intern**

This would involve shifting all the tests (units and integration) to Intern. We will do this using RequireJS's AMD loader and intern's qunit [module](#). This would also involve writing grunt tasks for testing via intern(part of prototype task) and ensure all current tests pass.

- **Cleanup the tests**

Current tests need to be cleaned up and refactored to new QUnit api. We can get the info of what all needs to be upgraded via [QUnit Migrate](#). Major example of tangled up tests is [navigation](#) tests. Tests also need to be written in proper way, for e.g. Use of promises instead of setTimeout. Proper assertions are to be placed everywhere and put a check of what is being asserted, for e.g. class checks in tests check if a class is equal to undefined, instead they should check if component has that class.

- **Make test pass on all supported browsers**

This part involves tweaking the api and tests to make them pass in all the supported browsers. At jQuery, we need to support many browsers including Internet Explorer (culprit most of the time) so it should be ensured that test pass on all browsers. Many tests assume some level of support and thus break, this part will handle all those tests.

Expected Results: A new test suite using the intern and passing on all supported devices via BrowserStack.

Timeline

No.	Timespan	Task
1.	April 27th - May 12th	Familiarising myself with the mentors, codebase and my project. Since I am currently actively participating with community, this won't take much time
2.	May 13th - May 21st	Upgrade to Intern (Configuration and grunt tasks)
3.	May 22nd - June 3rd	Upgrade to Intern (Port tests to intern)
4.	June 4th - June 10th	Cleanup and Refactor (Upgrade to QUnit 2.0 wherever necessary)
5.	June 10th - June 26th	Cleanup and Refactor (Part 1 - Unit Tests)
6.	June 26th	Midterm Evaluation
7.	June 27th - July 10th	Cleanup and Refactor (Part 2 - Integration Tests)
8.	July 11th - July 30th	Pass Tests on all supported browsers
9.	July 31st - August 11th	Write Integration Tests
10.	August 11th- August 17th	Wrap up everything
11.	August 28th	Final Evaluation

Details on Timeline:

- **Task 1:** During this time period I will get to know my mentors better and build up more strong bond with the community. I will dive deep into the code and learn all the ins and outs of the code with help of mentors.
- **Task 2:** During this task I will configure intern properly for use with project and write more grunt tasks (except the one in prototype) to automate the whole process of testing.
- **Task 3:** This task will involve the porting of current test suite to intern using RequireJS AMD and Intern's QUnit module. This would require wrapping all the tests mainly as a requirejs module.
- **Task 4:** Cleanup and refactor tests to use QUnit 2.0 API as it is going to be released soon. This tasks will mainly use [QUnit Migrate](#) repo which tells which tests need to be updated.
- **Task 5:** This phase will involve cleanup of unit tests to use proper assertions and promises everywhere. Code will be properly refactored to use proper equations at right places. Ambiguous comparisons and tests will be refactored to better ones.
- **Task 6:** Code refactored and cleaned up till this task's date will be submitted to Google.
- **Task 7:** Task involves cleanup of integration tests on the lines of unit tests.
- **Task 8:** Passing tests on all browsers is necessary so as to assure full support. Care will be taken of the tests that require extra support and will be fixed as suitable. Browserstack will be used to run tests. During this phase, aim would be to tweak tasks to pass on all supported browsers.
- **Task 9:** Currently integration tests in jQuery mobile repo are badly crafted and look hackish. Intern provides a simple API to simulate the real browser

environments. Thus I will aim to write simple and understandable functional tests to replace the much of the current integration tests.

- **Task 10**: Complete up the project and write whatever documentation and wikis are necessary. I will try to brush up code as much during this period.
- **Task 11**: Patches and diffs will be submitted to Google for review.

Additional Information

The most recent version of this proposal would be always available [here](#). Github will be used for hosting all the code at this [fork](#).

Participation

For me, It was and will be always - Sharing is Caring. This has always helped me to get well with the Open Source community. All source code I write will be published to my Github repository. I always try to stay live in IRC, and am regular in replying to emails, so it helps me to blend in the community. I am mostly available on #jquery-dev #jquery-ui-dev during my working hours.

About Me

I am a 20 year old, third year student currently enrolled in Electrical Engineering (IV Year Course) at [IIT Roorkee](#). I participated in GSoC last year with MediaWiki.

I developed a passion for programming and web development in my freshman year. I am regularly contributing to OpenSource community since then.

I am Joint Secretary of [SDSLabs](#) at IIT Roorkee. I am currently proficient in many languages including JavaScript, PHP, Python, Ruby, Go and many more.

I open source all my projects that I do individually so that the mass can gain something from it. I have been developing apps regularly at SDS Labs, we code late night at our lab and we all enjoy it.

I am having summer vacations from April end to July mid so I think I would be able to complete my project in time.

Availability

- Work hours per week
I usually work between 4:00 p.m. to 11:00 p.m. on weekdays and 11:00 a.m. to 11:00 p.m. on weekends, rest time is spent in usual exception of studies.
- Other commitments between May and August 2013
No
- Do you plan to apply for other internships during the summer?
No
- I feel comfortable discussing issues in the public developer forum
Yes
- I feel comfortable posting a weekly update to the public developer forum
Yes

Why jQuery?

I have been using jQuery ever since I remember since it is very popular JS library for beginners. That is why I have a natural interest to dive into jQuery's code and contribute back to it. Keeping all that in mind I selected jQuery for GSoC this year.